

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
 Nov – Dec 2015

Max. Marks: 100

Class: B.E

Name of the Course: Soft Computing

Branch: Computer

Course Code: UCEE705

Duration: 3 Hrs

Semester: VII

Instructions:

- (1) All Questions are Compulsory.
- (2) Assume suitable data if necessary.

Question No.		Max. Marks																								
Q. 1 (A)	Design a suitable Fuzzy logic controller for a subway train approaching or leaving a station .The inputs are distance from the station and speed of the train. The output is the amount of brake power applied. Design the Fuzzy logic controller using 4 descriptors for input and output variables	10																								
Q. 1 (B)	Describe the architecture of Mamdani type FIS in detail. Compare and contrast it with Sugeno FIS. OR Explain Fuzzy operations and Fuzzy relations with suitable examples.	08																								
Q. 1 (C)	Let $A = \{a_1, a_2\}$, $B = \{b_1, b_2, b_3\}$, $C = \{c_1, c_2\}$. Let R be a relation from A to B defining by matrix – <table border="1"><tr><td>R</td><td>b1</td><td>b2</td><td>b3</td></tr><tr><td>a1</td><td>0.5</td><td>0.6</td><td>0.2</td></tr><tr><td>a2</td><td>0.7</td><td>0.8</td><td>0.2</td></tr></table> Let S be a relation from B to C defined by matrix – <table border="1"><tr><td>S</td><td>c1</td><td>c2</td></tr><tr><td>b1</td><td>0.3</td><td>0.7</td></tr><tr><td>b2</td><td>0.5</td><td>0.8</td></tr><tr><td>b3</td><td>1</td><td>1</td></tr></table> Find 1. Max – min composition of R and S 2. Max – product composition of R and S	R	b1	b2	b3	a1	0.5	0.6	0.2	a2	0.7	0.8	0.2	S	c1	c2	b1	0.3	0.7	b2	0.5	0.8	b3	1	1	12
R	b1	b2	b3																							
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Q. 2 (A)	Explain Kohonen Self Organizing Map with supporting Diagrams. OR What is the significance of the learning in neural network? Compare the different types of learning algorithm.	10
Q. 2 (B)	Explain Error Back Propagation Training Algorithm with the help of a neat and detailed flowchart.	10
Q. 2 (C)	Determine the weights after one iteration for Perceptron learning rule for a single neuron network with the learning constant $c = 0.5$, the initial weight vector $W = [1 \ -1 \ 0.5]^T$ and the training input vectors are $X_1 = [1 \ -2 \ -1]^T$, $X_2 = [0 \ 1.5 \ -1]^T$ and $X_3 = [-1 \ 1 \ -1]^T$. And the desirable outputs are $d_1 = 0$, $d_2 = 1$, $d_3 = 1$. Consider bias.	10
Q. 3 (A)	Explain Genetic Algorithm with the help of flowchart. OR Explain classical newton's method of derivative based optimization techniques.	10
Q. 3 (B)	Give the basic steps involved in Downhill simplex search method of derivative free optimization.	10
Q. 4	Draw ANFIS for Sugeno fuzzy model and explain in detail. OR Explain Hybrid learning algorithm.	10
Q. 5	Write short note on ANY ONE a) Printed Character Recognition b) Soft Computing for Color Recipe Prediction	10